

DIFFERENTIAL CALCULUS

Basics

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx}$$

where $h = \Delta x, \Delta y = f(x+h) - f(x) = f(x+\Delta x) - f(x)$ provided the limits exists.

The differential of $y = f(x)$ is defined by
 $dy = f'(x)dx$

where $dx = \Delta x$.

Indeterminate Forms

$$1^\infty, 0^0, \infty - \infty, \frac{0}{0}, \frac{\infty}{\infty}, \infty^0, 0 \cdot \infty$$

Via 3 Point Test $\rightarrow$ approaching

[CALC] function

L'HOSPITAL'S RULE

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow a} \frac{f''(x)}{g''(x)} = \dots$$

Rate of Change of Slope

- $y'' > 0$, rotation of tangents is counterclockwise, concave upward
- $y'' < 0$, rotation of tangents is clockwise, concave downward
- $y' = \infty$ (vertical line)

Maxima & Minima

- if $y' = 0, y'' > 0$, critical point is minima

- if $y' = 0, y'' < 0$, critical point is maxima
- if $y' = 0, y'' = 0$, point is not an extremum

POINT	y'	y''	OPENS
max	0	-	down
min	0	+	up
inflection	$\neq 0$	0	--

Point of Inflection

$$y'' = 0$$

Slope of Polar Curves

$$\text{slope} = \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

CALTECH: Second Derivative

$$\textcircled{1} \frac{d}{dx} [f(x)] dx \Big|_{x=A} \rightarrow \textcircled{A}$$

$$\textcircled{2} \frac{d}{dx} [f(x)] dx \Big|_{x=B} \rightarrow \textcircled{B}$$

$$\textcircled{3} \frac{\textcircled{B} - \textcircled{A}}{B - A} = y'' \text{ @ } x = A$$

Curvature of a Curve

$$k = \frac{y''}{[1 + (y')^2]^{3/2}}$$

Radius of a Curvature

$$R = \left| \frac{1}{k} \right| = \left| \frac{[1 + (y')^2]^{3/2}}{y''} \right|$$

Center of Curvature (α, β)

$$\alpha = x - \frac{y' [1 + (y')^2]}{y''}$$

$$\beta = y + \frac{1 + (y')^2}{y''}$$

Algebraic Functions

$$1. \frac{dc}{dx} = 0$$

$$2. \frac{d}{dx} (u + v) = \frac{du}{dx} + \frac{dv}{dx}$$

$$3. \frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$4. \frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$5. \frac{d}{dx} (cv) = c \frac{dv}{dx}$$

$$6. \frac{d}{dx} \left(\frac{1}{v} \right) = - \frac{\frac{dv}{dx}}{v^2}$$

$$7. \frac{d}{dx} (u^n) = nu^{n-1} \frac{du}{dx}$$

$$8. \frac{d}{dx} \sqrt{u} = \frac{\frac{du}{dx}}{2\sqrt{u}}; u \neq 0$$

$$9. \frac{d}{dx} \left(\frac{c}{v} \right) = \frac{-c \frac{dv}{dx}}{v^2}$$

Trigonometric Functions

- $\frac{d}{dx} \sin u = \cos u \frac{du}{dx}$
- $\frac{d}{dx} \cos u = -\sin u \frac{du}{dx}$
- $\frac{d}{dx} \tan u = \sec^2 u \frac{du}{dx}$
- $\frac{d}{dx} \cot u = -\csc^2 u \frac{du}{dx}$
- $\frac{d}{dx} \sec u = \sec u \tan u \frac{du}{dx}$
- $\frac{d}{dx} \csc u = -\csc u \cot u \frac{du}{dx}$

Inverse Trigonometric Functions

$$1. \frac{d(\arcsin u)}{dx} = \frac{du/dx}{\sqrt{1-u^2}}$$

$$2. \frac{d(\arccos u)}{dx} = \frac{-du/dx}{\sqrt{1-u^2}}$$

$$3. \frac{d(\arctan u)}{dx} = \frac{du/dx}{1+u^2}$$

$$4. \frac{d(\operatorname{arccot} u)}{dx} = -\frac{du/dx}{1+u^2}$$

$$5. \frac{d(\operatorname{arcsec} u)}{dx} = \frac{du/dx}{u\sqrt{u^2-1}}$$

$$6. \frac{d(\operatorname{arccsc} u)}{dx} = -\frac{du/dx}{u\sqrt{u^2-1}}$$

Logarithmic & Exponential Functions

- $\frac{d}{dx} \ln u = \frac{du}{u}; u \neq 0$
- $\frac{d}{dx} \log_a u = \frac{du}{u \ln a}$
- $\frac{d}{dx} \log_{10} u = M \frac{du}{u} = 0.43429 \frac{du}{u}$ where $M = \log_{10} e = 0.43429$
- $\frac{d}{dx} a^u = a^u \ln a \frac{du}{dx}$
- $\frac{d}{dx} e^u = e^u \frac{du}{dx}$
- $\frac{d}{dx} (u^v) = v u^{v-1} \frac{du}{dx} + u^v \ln u \frac{dv}{dx}$

$$1) \frac{d(\log_a u)}{dx} = \log_a e \frac{du/dx}{u}$$

Hyperbolic Functions

$$1) \frac{d(\sinh u)}{dx} = \cosh u \frac{du}{dx}$$

$$2) \frac{d(\cosh u)}{dx} = \sinh u \frac{du}{dx}$$

$$3) \frac{d(\tanh u)}{dx} = \operatorname{sech}^2 u \frac{du}{dx}$$

$$4) \frac{d(\coth u)}{dx} = -\operatorname{csch}^2 u \frac{du}{dx}$$

$$5) \frac{d(\operatorname{sech} u)}{dx} = -\operatorname{sech} u \tanh u \frac{du}{dx}$$

$$6) \frac{d(\operatorname{csch} u)}{dx} = -\operatorname{csch} u \coth u \frac{du}{dx}$$

Other Formulas

SLOPE OF THE CURVE

The slope of the curve $y = f(x)$ at any point is identical to the derivative of the function $\frac{dy}{dx}$ or y' .

Slope at any point $= y' = \frac{dy}{dx}$

RATE OF CHANGE

The derivative of a function is identical to its rate of change. Thus, the rate of change of the volume (V) of a sphere with respect to its radius (r) is dV/dr .

CURVATURE AND RADIUS OF CURVATURE

CURVATURE

Curvature refers to the rate of change of the direction of the curve. Thus, a circle with a smaller radius has great curvature, or is sharply curved.

The curvature, k , of the curve $y = f(x)$ is:

$$k = \frac{y''}{[1 + (y')^2]^{3/2}}$$

where y'' is the second derivative of the function, and $|y'|$ is the absolute or positive value of y' .

RADIUS OF CURVATURE

The radius of curvature, ρ , is the reciprocal of the curvature, k , or:

$$\rho = \frac{1}{k} = \frac{[1 + (y')^2]^{3/2}}{|y'|}$$

Circle of Curvature

At any point on a curve $y = f(x)$, where y' and y'' exist and $y'' \neq 0$, there is associated with the curve a circle, which is called the circle of curvature with the following equation:

$$(x - h)^2 + (y - k)^2 = \rho^2$$

The center (h, k) and radius ρ of the curve is:

$$h = x - \frac{y'[1 + (y')^2]}{y''}; \quad k = y + \frac{1 + (y')^2}{y''}$$

$$\rho = \frac{[1 + (y')^2]^{3/2}}{|y'|}$$

Common Maxima-minima Formulae

- Maximum/minimum points: $y' = 0$
- maxima: $y'' < 0$ (concave downward)

- minima: $y'' > 0$ (concave upward)
- point of inflection: $y'' = 0$

INTEGRAL CALCULUS

Indefinite Integral

Algebraic, Exponential, & Logarithm

$$\int a \, du = a \int du = au + C$$

$$\int a^u \, du = \frac{a^u}{\ln a} + C$$

$$\int u^n \, du = \frac{1}{n+1} u^{n+1} + C \text{ for } n \neq -1$$

$$\int e^u \, du = e^u + C$$

$$\int u^{-1} \, du = \int \frac{du}{u} = \ln u + C$$

$$\int \ln u \, du = u \ln u - u + C$$

Trigonometric Functions

$$\int \sin u \, du = -\cos u + C$$

$$\int \sec^2 u \, du = \tan u + C$$

$$\int \cos u \, du = \sin u + C$$

$$\int \csc^2 u \, du = -\cot u + C$$

$$\int \tan u \, du = \ln |\sec u| + C$$

$$\int \sec u \tan u \, du = \sec u + C$$

$$\int \cot u \, du = \ln |\sin u| + C$$

$$\int \csc u \cot u \, du = -\csc u + C$$

$$\int \sec u \, du = \ln |\sec u + \tan u| + C$$

$$\int \csc u \, du = \ln |\csc u - \cot u| + C$$

Inverse Trigonometric Functions

$$\int \frac{du}{\sqrt{a^2 - u^2}} = \arcsin \frac{u}{a} + C$$

$$\int \frac{du}{a^2 - u^2} = \frac{1}{a} \arctan \frac{u}{a} + C$$

$$\int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \operatorname{arcsec} \frac{u}{a} + C$$

$$\int \arcsin u \, du = u \arcsin u + \sqrt{1 - u^2} + C$$

$$\int \arctan u \, du = u \arctan u - \ln \sqrt{1 + u^2} + C$$

Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

Trigonometric Substitution

when $a^2 - u^2$ is involved, try $u = a \sin \theta$

when $a^2 + u^2$ is involved, try $u = a \tan \theta$

when $u^2 - a^2$ is involved, try $u = a \sec \theta$

Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

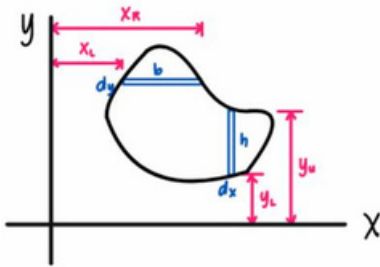
Wallis Formula

$\alpha = \frac{\pi}{2}$, when both m & n are even
 $\alpha = 1$, if otherwise

$$\int_0^{\pi/2} \sin^m \theta \cos^n \theta \, d\theta = \frac{[(m-1)(m-3)\dots(2 \text{ or } 1)][(n-1)(n-3)\dots(2 \text{ or } 1)]}{(m+n)(m+n-2)\dots(2 \text{ or } 1)} \cdot \alpha$$

$$\int \csc u \, du = \ln |\csc u - \cot u| + C$$

Plane Areas (Rectangular Coordinates)



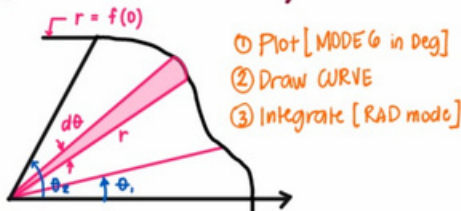
Horizontal Strip

$$A = \int_{y_1}^{y_2} b \, dy = \int_{y_1}^{y_2} (x_R - x_L) \, dy$$

Vertical Strip

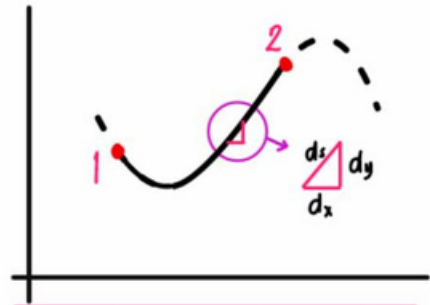
$$A = \int_{x_1}^{x_2} h \, dx = \int_{x_1}^{x_2} (y_u - y_L) \, dx$$

Plane Areas (Polar Coordinates)



$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} r^2 \, d\theta$$

Length of Arc

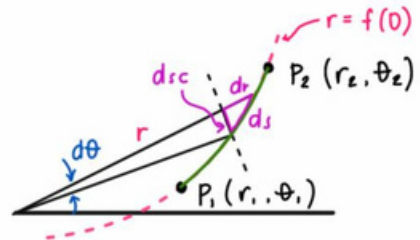


$$ds^2 = dx^2 + dy^2$$

$$S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

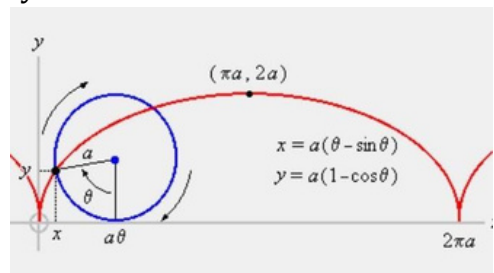
$$S = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy$$

Length of Polar Curve



$$S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} \, d\theta$$

Cycloid



- $x = a(\theta - \sin \theta)$
- $y = a(1 - \cos \theta)$
o a = radius of rolling circle

$\theta =$ <i>angle of rotation</i>
TRANSPORTATION
Number of Cars Expected on the Ramp $E = \frac{p}{1-p} \left(\frac{1 + (N+1)p^N + Np^{N+1}}{1-p^{N+1}} \right)$ • p = arrival rate / permissible cars • N = capacity of ramp